Validation of numerical results obtained on MATLAB

Verification of L,C values

Solved Example 1:

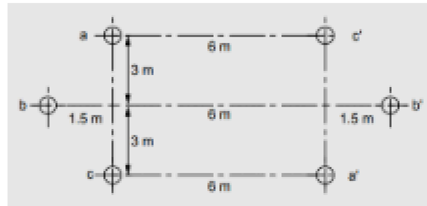


Fig. 1.22.3

Solution : The stranded conductor is specified as no. of strands and diameter of each strand 7/4.75 mm hard drawn copper conductors are there.

d = Diameter of each strand = 4.75 mm.

Total diameter of stranded conductor, dc = (2n + 1)d where n = No. of layers in which the strands are arranged around central strands.

If no. of layers are not specified then the no. of layers n are given by,

$$X = 3n^2 + 3n + 1$$

where X = No. of strands = 7

$$7 = 3n^2 + 3n + 1$$

$$\therefore \quad 3n^2 + 3n - 6 = 0$$

$$\therefore \quad n^2 + n - 2 = 0$$

$$\therefore \quad (n+2)(n-1) = 0$$

$$\therefore \quad n = -2,\ n = 1$$

Neglecting negative root as layers cannot be negative.

$$\therefore \quad n = 1$$

$$\therefore \quad d_c = (2n+1)\,d = [2(1)+1]\,d = 3d = 3 \times 4.75 = 14.25 \text{ mm}$$

$$d_c = 0.01425 \text{ m}$$

Radius of each conductor,

$$r = \frac{d_c}{2} = 7.125 \times 10^{-3} \text{ m}$$

$$r' = 0.7788\, r$$

$$= 0.7788 \times 7.125 \times 10^{-3} = 5.5489 \times 10^{-3} \text{ m}$$

$$= 0.00555 \text{ m}$$

$$D_{ab} = \sqrt{(3)^2 + (1.5)^2} = 3.3541 \text{ m} = D_{bc'} \quad D_{bb'} = 9 \text{ m}$$

$$D_{ab'} = \sqrt{(3)^2 + (7.5)^2} = 8.077 \text{ m} = D_{ba'} = D_{bc'},\ D_{ca} = D_{ac} = 6 \text{ m},\ D_{ca'} = 6 \text{ m}$$

$$D_{aa'} = \sqrt{(6)^2 + (6)^2} = 8.4852 \text{ m} = D_{cc'}$$

D_s is the equivalent GMD of one phase which is given by,

$$D_s = \sqrt[4]{D_{s1} \cdot D_{s2} \cdot D_{s3}}$$

$$D_{s1} = \sqrt[4]{D_{aa} D_{aa'} \cdot D_{a'a} D_{a'a'}} = \sqrt[4]{(0.00555)(8.4852)(8.4852)(0.00555)}$$

$$= 0.2170 \text{ m}$$

$$D_{s2} = \sqrt[4]{D_{bb} D_{bb'} D_{b'b} D_{b'b'}} = \sqrt[4]{(0.00555)(9)(9)(0.00555)}$$

$$= 0.2234 \text{ m}$$

$$D_{s3} = \sqrt[4]{D_{cc} D_{cc'} D_{c'c} D_{c'c'}} = \sqrt[4]{(0.00555)(8.4852)(8.4852)(0.00555)}$$

$$= 0.2170 \text{ m}$$

$$D_s = \sqrt[3]{(0.2170)(0.2234)(0.2170)} = 0.2191 \text{ m}$$

$$D_m = \sqrt[3]{(D_{AB} \cdot D_{BC} \cdot D_{CA}}$$

$$D_{AB} = \sqrt[4]{D_{ab} \cdot D_{ab'} \cdot D_{ba} D_{ba'}} = \sqrt[4]{(3.3541)(8.077)(3.3541)(8.077)}$$

$$= 5.2049 \text{ m} = D_{BC}$$

$$D_{CA} = \sqrt[4]{D_{ca} D_{ca'} \cdot D_{ac} D_{ac'}} = \sqrt[4]{(6)(6)(6)(6)} = 6 \text{ m}$$

$$D_m = \sqrt[3]{(5.2049)(5.2049)(6)} = 5.4574 \text{ m}$$

$$\text{Inductance per phase per km} = 2 \times 10^{-4} \ln\left(\frac{D_m}{D_s}\right)$$

$$= 2 \times 10^{-4} \ln\left(\frac{5.4574}{0.2191}\right)$$

$$= 6.4304 \times 10^{-4} \text{ H}$$

Matlab Result 1:

```
Enter operating temperature (°C): 67
Enter line length (km): 45
Enter unit for diameters/spacings (m / cm / mm / in / ft) : m
Enter strand diameter (m): 4.75*10^-3
Enter number of strands in bundle: 7

Total line resistance at 67.0°C = 5.65418 Ω (total line)
Number of subconductor bundles per phase (1-12): 1
Number of circuits (1/2): 2
Enter the horizontal phase spacing x (m): 6
Enter the vertical phase spacing y (m): 6
Enter the middle phase displacement spacing z (m): 1.5
GMR = 2.191297e-01 m
GMD = 5.457653e+00 m
L = 6.430221e-04 H/km
C = 1.800341e-08 F/km
```

Solved Example 2:

Example 1.19.5 *A 50 Hz overhead transmission line consisting of 3 conductors each of diameter 1.24 cm and spaced 2m apart. Calculate the inductance per phase per km for the following arrangement between conductors :*

1) Equilateral spacing 2) Horizontal spacing.

Assume transposed line.

Solution : First let's consider the equilateral spacing of conductors.

D = Distance between the conductors = 2 m.

Radius of conductor = Diameter of conductor / 2

$$= \frac{1.24 \times 10^{-2}}{2} = 6.2 \times 10^{-3} \text{ m}$$

$$r' = 0.7788 \times r$$

$$= 0.7788 \times 6.2 \times 10^{-3} = 4.8285 \times 10^{-3} \text{ m}$$

Inductance per phase per km is given by,

$$L_A = 2 \times 10^{-4} \, ln\left(\frac{D}{r'}\right) \text{H/km}$$

$$= = 2 \times 10^{-4} \, ln\left[\frac{2}{4.8285 \times 10^{-3}}\right]$$

$$= = 1.2052 \times 10^{-3} \text{ H/km}$$

$$\therefore \quad \mathbf{L_A = 1.2052 \text{ mH/km}}$$

tal

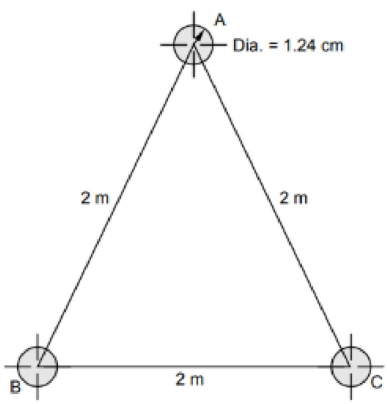


Fig. 1.19.7 (a)

Matlab Result 2:

```
Enter operating temperature (°C): 75
Enter line length (km): 34
Enter unit for diameters/spacings (m / cm / mm / in / ft) : cm
Enter conductor diameter (cm): 1.24
Number of subconductor bundles per phase (1-12): 1
Number of circuits (1-4): 1
Enter the phase spacing x (cm): 200
GMR = 4.828560e-03 m
GMD = 2.000000e+00 m
L = 1.205271e-03 H/km
C = 9.631077e-09 F/km
Inductance = 1.205271e-03 H/km
Capacitance = 9.631077e-09 F/km
```

Solved Example 3:

Example 1.19.6 *Determine the inductance of a 3 phase, line operating at 50 Hz and the conductors are arranged as shown in Fig. 1.19.8. The conductor diameter is 0.7 cm.*

AU : May-05, May-17, Marks 8

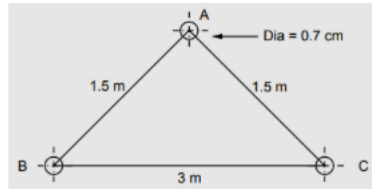


Fig. 1.19.8

Solution :

Solution : Radius of conductor $= \frac{0.7}{2} = 0.35 \text{ cm} = 3.5 \times 10^{-3}\text{m}$

$$r' = 0.7788 \times 3.5 \times 10^{-3} = 2.7258 \times 10^{-3}\text{m}$$

$$D_{AB} = 1.5 \text{ m}$$

$$D_{BC} = 3 \text{ m}$$

$$D_{CA} = 1.5 \text{ m}$$

$$D_{eq} = \sqrt[3]{D_{AB} \cdot D_{BC} \cdot D_{CA}} = \sqrt[3]{(1.5)(3)(1.5)} = 1.8898 \text{ m}$$

Inductance per phase,

$$L_A = 2 \times 10^{-7}\, ln\left[\frac{D_{eq}}{r'}\right] \text{H/m} = 2 \times 10^{-7}\, ln\left[\frac{1.8898}{2.7258 \times 10^{-3}}\right]$$

$$= 1.3083 \times 10^{-6} \text{ H/m}$$

$$= 1.3083 \times 10^{-3} \text{ H/km}$$

$$\therefore \quad L_A = \mathbf{1.31 \text{ mH/km}}$$

Matlab Result 3:

```
Enter operating temperature (°C): 75
Enter line length (km): 12
Enter unit for diameters/spacings (m / cm / mm / in / ft) : cm
Enter conductor diameter (cm): 0.7
Number of subconductor bundles per phase (1-12): 1
Number of circuits (1-4): 1
Enter the phase spacing x (cm): 150
Enter the phase spacing x (cm): 300
Enter the phase spacing x (cm): 150
AB = 1.500 m, BC = 3.000 m, CA = 1.500 m
GMR = 2.725800e-03 m
GMD = 1.889882e+00 m
L = 1.308301e-03 H/km
C = 8.842477e-09 F/km
Inductance = 1.308301e-03 H/km
Capacitance = 8.842477e-09 F/km
```

Solved Example 4:

$D_{AB} = 2$ m, $D_{BC} = 2$ m; $D_{CA} = 4$ m

We have, $r' = 4.8285 \times 10^{-3}$ m

$$D_{eq} = \sqrt[3]{D_{AB}\, D_{BC}\, D_{CA}} \quad \sqrt[3]{(2)\,(2)\,(4)}$$

$$= \; = \sqrt[3]{16} = 2.5198 \text{ m}$$

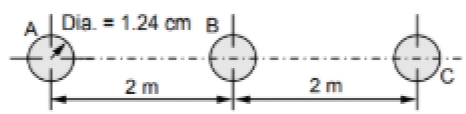


Fig. 1.19.7 (b)

Inductance per phase per km is given as,

$$L_A = 2 \times 10^{-4}\, ln\left(\frac{D_{eq}}{r'}\right) \text{H/km} = 2 \times 10^{-4}\, ln\left(\frac{2.5198}{4.8285 \times 10^{-3}}\right) = 1.2514 \times 10^{-3} \text{ H/km}$$

$$\therefore \quad \mathbf{L_A = 1.2514 \text{ mH/km}}$$

Matlab Result 4:

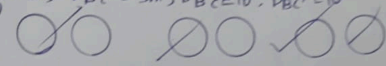
```
Enter operating temperature (°C): 75
Enter line length (km): 34
Enter unit for diameters/spacings (m / cm / mm / in / ft) : cm
Enter conductor diameter (cm): 1.24
Number of subconductor bundles per phase (1-12): 1
Number of circuits (1-4): 1
Enter the phase spacing x (cm): 200
GMR = 4.828560e-03 m
GMD = 2.519842e+00 m
L = 1.251481e-03 H/km
C = 9.260659e-09 F/km
Inductance = 1.251481e-03 H/km
Capacitance = 9.260659e-09 F/km
```

Solved Example 5: (CAT1 Qn)

4)

a) $D_{CA} = 6$; $D_{C'A'} = 6$; $D_{A'C} = 13$; $D_{C'A} = 13$

$D_{BC} = 3$; $D_{B'C'} = 3m$; $D_{B'C} = 10$; $D_{BC'} = 10$

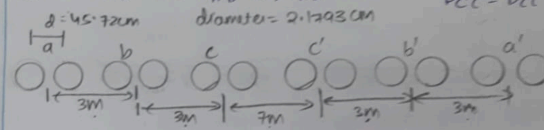


$D_{A'A} = D_{AA'} = 19m.$

$D_{B'B} = D_{BB'} = 13m.$

$D_{C'C} = D_{CC'} = 7m$

$D_{AB} = 3$; $D_{A'B'} = 3$; $D_{A'B} = 16$; $D_{AB'} = 16$.



$$\text{Radius} = \frac{2.1793}{2} \times 10^{-2}\ m = 0.01089\ m$$

$$R' = GMR = 0.7788 \times \text{Radius} = 8.4861 \times 10^{-3}$$

$$D_{AA} = \sqrt{R'd} = \sqrt{8.4861 \times 10^{-3} \times 45.72 \times 10^{-2}} = 0.0622m$$

$$D_{AA} = D_{A'A'} = D_{BB} = D_{B'B'} = D_{CC} = D_{C'C'}$$

$$D_{SA} = \sqrt[4]{D_{AA}\, D_{A'A'}\, D_{A'A}\, D_{AA'}} = \sqrt[4]{0.0622^2 \times 19^2} = 1.0871m$$

$$D_{SB} = \sqrt[4]{D_{BB}\, D_{B'B'}\, D_{BB'}\, D_{B'B}} = \sqrt[4]{0.0622^2 \times 13^2} = 0.899\ m.$$

$$D_{SC} = \sqrt[4]{D_{CC}\, D_{C'C'}\, D_{CC'}\, D_{C'C}} = \sqrt[4]{0.0622^2 \times 7^2} = 0.659\ m$$

$$D_S = \sqrt[3]{D_{SA}\, D_{SB}\, D_{SC}} = \sqrt[3]{1.087 \times 0.899 \times 0.659} = 0.8635m$$

$$D_{AB} = \sqrt[4]{D_{AB}\, D_{A'B'}\, D_{AB'}\, D_{A'B}} = \sqrt[4]{16^2 \times 3^2} = 6.9282m.$$

$$D_{BC} = \sqrt[4]{D_{BC}\, D_{BC'}\, D_{B'C}\, D_{B'C'}} = \sqrt[4]{10^2 \times 3^2} = 5.477m$$

$$D_{CA} = \sqrt[4]{D_{CA}\, D_{CA'}\, D_{C'A}\, D_{C'A'}} = \sqrt[4]{13^2 \times 6^2} = 8.8317m$$

$$D_m = \sqrt[3]{D_{AB}\, D_{BC}\, D_{CA}} = \sqrt[3]{6.9282 \times 5.477 \times 8.831} = 6.9458m$$

$$l = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) = 2 \times 10^{-7} \ln\left(\frac{6.9458}{0.8635}\right) = 0.4169\ mH/km$$

5

Matlab Result 5

```
Enter operating temperature (°C): 45
Enter line length (km): 45
Enter unit for diameters/spacings (m / cm / mm / in / ft) : m
Enter conductor overall diameter (m): 2.1793*10^-2

Total line resistance at 45.0°C = 3.44630 Ω (total line)
Number of subconductor bundles per phase (1-12): 2
Number of circuits (1/2): 2
Enter the internal bundle spacing b (m): 45.72*10^-2
Enter the phase spacing x (m): 3
Enter the phase spacing u (m): 7
GMR = 8.646439e-01 m
GMD = 6.946129e+00 m
L = 4.167244e-04 H/km
C = 2.752556e-08 F/km
```

Solved Example 6: (CAT1 Qn)

b) d = spacing within bundle = 45·72cm

Diameter = 2·1793cm.

$$\text{Radius} = \frac{2.1793}{2} \times 10^{-2} = 0.010896\ m.$$

$$D_{AA} = \sqrt[3]{\text{Radius} \times d^2} = 0.13157m = D_{BB} = D_{CC} = \dots = D_{C'C'}$$

$$D_{SA} = \sqrt[4]{D_{AA} \times D_{A'A'}\, D_{AA'}\, D_{A'A}} = \sqrt[4]{(0.13157)^2 \times 19^2} = 1.581m$$

$$D_{SB} = \sqrt[4]{D_{BB} \times D_{B'B'}\, D_{BB'}\, D_{B'B}} = \sqrt[4]{(0.13157)^2 \times 13^2} = 1.3078m$$

$$D_{SC} = \sqrt[4]{D_{CC} \times D_{C'C'}\, D_{CC'}\, D_{C'C}} = \sqrt[4]{(0.13157)^2 \times 7^2} = 0.9596m$$

$$D_S = \sqrt[3]{D_{SA}\, D_{SB}\, D_{SC}} = \sqrt[3]{1.581 \times 1.3078 \times 0.9596} = 1.2565m$$

$D_{AB} = 6.9282m$

$D_{BC} = 5.477m$

$D_{CA} = 8.8317\ m$

→ Same values & calc^n as prev., as spacing b/w phases are same only bundle spacing & config^n is different.

$$\therefore D_m = \sqrt[3]{D_{AB}\, D_{BC}\, D_{CA}}$$

$\Rightarrow 6.9458\ m$

$$C = \frac{2\pi\varepsilon_0}{\ln\left(\frac{D_m}{D_s}\right)} = \frac{2\pi\varepsilon_0}{\ln\left(\frac{6.9458}{1.2565}\right)} = 3.25 \times 10^{-11}\ F/m$$

$\Rightarrow 32.52\ pF/m$

$\Rightarrow 32.52\ nF/km$

```
Enter operating temperature (°C): 45
Enter line length (km): 45
Enter unit for diameters/spacings (m / cm / mm / in / ft) : m
Enter conductor overall diameter (m): 2.1793*10^-2

Total line resistance at 45.0°C = 3.44630 Ω (total line)
Number of subconductor bundles per phase (1-12): 3
Number of circuits (1/2): 2
Enter the internal bundle spacing b (m): 45.72*10^-2
Enter the phase spacing x (m): 3
Enter the phase spacing y (m): 3
Enter the phase spacing u (m): 7
GMR = 1.205369e+00 m
GMD = 6.946129e+00 m
L = 3.502797e-04 H/km
C = 3.253873e-08 F/km
```

Solved Example 7:

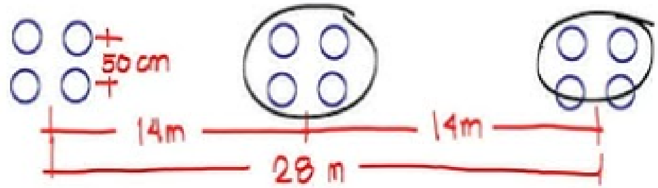
A single circuit three phase transmission line is composed of four ACSR conductor per phase with horizontal configuration. The conductors are spaced 50 cm in the bundle and the spacing between adjacent phases is 14 m. Find the inductance per km. The radius of each conductor in the bundle is 1.725cm. Ans: L=0.876 mH/km

3ϕ! Bundle of 4

$$L = 2\times10^{-7} \ln\left[\frac{GMD}{GMR}\right]$$

$$GMR = 1.09\sqrt[4]{(\text{self } GMR)(d_B)^3}$$

$r = 1.725$ cm



for GMD!

$$GMD = \sqrt[3]{(14m)(}$$

```
Enter operating temperature (°C): 45
Enter line length (km): 45
Enter unit for diameters/spacings (m / cm / mm / in / ft) : m
Enter conductor overall diameter (m): 1.725*2*10^-2

Total line resistance at 45.0°C = 0.89304 Ω (total line)
Number of subconductor bundles per phase (1-12): 4
Number of circuits (1/2): 1
Enter the internal bundle spacing b (m): 50*10^-2
Enter the phase spacing x (m): 14
GMR = 2.207547e-01 m
GMD = 1.763889e+01 m
L = 8.761619e-04 H/km
C = 1.288294e-08 F/km
```

Verification of Efficiency and Regulation

Question 1:

A single phase 50 Hz generator supplies an inductive load of 6 MW at 0.8 pf lagging by means of an overhead line 15 km long. The line resistance and inductance are 0.02 ohm/km and 0.85 mH/km. The voltage at the receiving end is 11 kV. Determine the sending end voltage and voltage regulation.

Solution : $P = 6$ MW, $\cos\phi_R = 0.8$ lag, $\phi_R = 36.86°$, 15 km

$$\therefore \quad Z = R + jX_L = (0.02 \times 15) + j\,(2\pi{\times}50 \times 0.85 \times 15 \times 10^{-3})$$

$$= 0.3 + j\,4\ \Omega = 4.0112\ \angle 85.711°\ \Omega, \quad V_R = 11\ \text{kV}$$

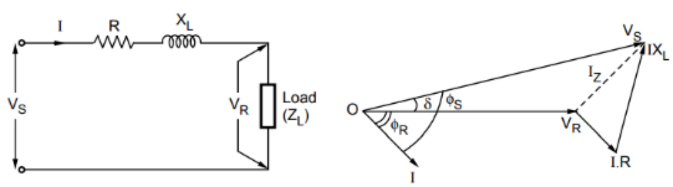


Fig. 2.5.5

$$I = \frac{P}{V_R \cos\phi_R} = \frac{6{\times}10^6}{11{\times}10^3{\times}0.8} = 681.818\ \text{A} \qquad \text{...Single phase}$$

$$\therefore \quad I = 681.818\ \angle -36.86°\ \text{A} = 545.525 - j\,409\ \text{A}$$

$$\therefore \quad \overline{V}_S = \overline{V}_R + \overline{I}\,\overline{Z} = [11{\times}10^3 \angle 0°] + [681.818 \angle -36.86°] {\times} 4.0112 \angle 85.711°$$

$$= 11 \times 10^3 + j\,0 + 1800 + j\,2059.388 = 12800 + j\,2059.388$$

$$= 12964.61\ \angle 9.14°\ \text{V}$$

$\therefore$ Sending end voltage = **12.9646 kV**

$$\%\,R = \frac{V_S - V_R}{V_R} \times 100 = \frac{12.9646 - 11}{11} \times 100 = \mathbf{17.86\ \%}$$

```
Enter frequency (Hz): 50
Enter line length (km): 15
Enter resistance per km (ohm/km): 0.02
Enter inductance per km (H/km): 0.85*10^-3
Enter capacitance per km (F/km): 0
Enter conductance per km (S/km) [enter 0 if not known]: 0

Line classification: Short Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 0

Enter Receiving End RMS Voltage (V): 11000
Enter load power (W): 6*10^6
Enter power factor (positive for lagging, negative for leading): 0.8

RESULTS FOR SHORT LINE MODEL
A = 1.0000 +0.0000j
B = 0.3000 +4.0055j
C = 0.0000 +0.0000j
D = 1.0000 +0.0000j

Receiving end current magnitude: 681.8182 A
Sending end current magnitude  : 681.8182 A

Sending end voltage magnitude   : 12967.27 V
Receiving end voltage magnitude : 11000.00 V
Voltage Regulation              : 17.88 %
Efficiency                      : 97.73 %
Total losses                    : 139462.8099 W
Sending Power: 6139462.8099 + j6362075.1911 VA
```

Question 2

Solution : Load p.f., $\cos \phi_R = 0.8$ lagging $\therefore \phi_R = 36.86°$

$$V_R = 11 \text{ kV}, \quad \text{Power delivered, } P_R = 5 \text{ MW} = 5 \times 10^6 \text{ W}$$

$$X_L = (2\pi f L) \times \text{Length of line} = (2\pi \times 50 \times 1.1 \times 10^{-3}) \times 15$$

$$= 5.1836\ \Omega$$

$$\text{Line losses} = 12\ \%\ \text{of power delivered}$$

$$= \left(\frac{12}{100}\right)(5 \times 10^6)$$

$$= 0.6 \times 10^6 \text{ W} = 0.6 \text{ MW}$$

Received end current, I_R is given as,

Power delivered,

$$P_R = \sqrt{3}\, V_R I_R \cos \phi_R$$

$$\therefore \quad I_R = \frac{P_R}{\sqrt{3}\, V_R \cos\phi_R} = \frac{5 \times 10^6}{\sqrt{3} \times 11 \times 10^3 \times 0.8} = 328.03 \text{ A}$$

$$\text{Total line losses} = 3 I_R^2 \cdot R = (3)\,(328.03)^2 \cdot R$$

$$\therefore \quad 0.6 \times 10^6 = (3)\,(328.03)^2 \cdot R$$

$$\therefore R = \frac{0.6 \times 10^6}{(3)\,(328.03)^2} = 1.85\ \Omega$$

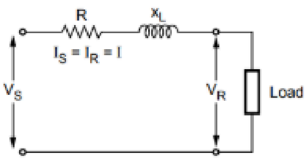


Fig. 2.7.2

$$I = I_S = I_R = 328.03 \angle -36.86° \text{ A}$$

$$\bar{V}_S = \bar{V}_R + \bar{I}\,\bar{Z} = \bar{V}_R + \bar{I}\,(R + jX_L)$$

$$\text{Let } \bar{V}_R = V_R \angle 0° = \frac{11000}{\sqrt{3}} \angle 0°$$

$$\therefore \quad \bar{V}_R = 6350.85 \angle 0° \text{ volts}$$

$$\bar{V}_S = [6350.85 \angle 0°] + [328.03 \angle -36.86°]\,[1.85 + j\,5.1836]$$

$$\therefore \quad \bar{V}_S = [6350.85 + j0] + [328.03 \angle -36.86°]\,[5.50 \angle 70.35°]$$

$$\therefore \quad \bar{V}_S = [6350.85 + j0] + [1804.165 \angle 33.49°]$$

$$= [6350.85 + j0] + [1504.64 + j\,995.52]$$

$$= (7855.49 + j995.52) = 7918.31 \angle 7.22° \text{ volts}$$

$$\therefore \quad V_S = 7.92 \text{ kV}$$

$$V_{S\text{ line}} = \sqrt{3} \cdot V_S = \sqrt{3} \times 7.92 = 13.71 \text{ kV}$$

$$\%\ \text{Voltage regulation} = \frac{V_S - V_R}{V_R} \times 100 = \frac{7918.31 - 6350.85}{6350.85} \times 100$$

$\therefore$ Voltage regulation = **24.68 %**

```
Enter frequency (Hz): 50
Enter line length (km): 15
Enter resistance per km (ohm/km): 0.12333
Enter inductance per km (H/km): 1.11*10^-3
Enter capacitance per km (F/km): 0
Enter conductance per km (S/km) [enter 0 if not known]: 0

Line classification: Short Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End RMS Voltage (V): 11000
Enter load power (W): 5*10^6
Enter power factor (positive for lagging, negative for leading): 0.8

RESULTS FOR SHORT LINE MODEL
A = 1.0000 +0.0000j
B = 1.8499 +5.2308j
C = 0.0000 +0.0000j
D = 1.0000 +0.0000j

Receiving end current magnitude: 328.0399 A
Sending end current magnitude  : 328.0399 A

Sending end voltage magnitude   : 7930.28 V
Receiving end voltage magnitude : 6350.85 V
Voltage Regulation              : 24.87 %
Efficiency                      : 89.33 %
Total losses                    : 199073.4762 W
Sending Power: 1865740.1429 + j1812882.2065 VA
```

Question 3

Solution : Total line length = 100 km

$$L = 1.21 \times 10^{-3}\ \text{H/km/phase}, \quad C = 0.00958\ \mu\text{F/km/phase}$$

$$\text{Total } R = 0.153 \times 100 = 15.3\ \Omega$$

$$\text{Total } X_L = (2\pi f L)\times 100 = (2\times\pi\times 50\times 1.21\times 10^{-3})(100) = 38\ \Omega$$

$$\text{Total } X_C = \frac{1}{(2\pi f C_{ph})\times 100} = \frac{1}{2\pi\times 50\times 0.00958\times 10^{-6}\times 100} = 3322.65\ \Omega$$

$$Z_C = 0 - jX_C\ ;\ Y_C = \frac{1}{Z_C} = \frac{1}{-jX_C} = j\left(\frac{1}{X_C}\right) = j\,3\times 10^{-4}\ ℧$$

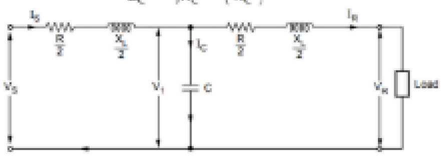


Fig. 2.8.7

$$\overline{V}_R = V_R\angle 0^\circ \text{ volts} = \frac{110\times 10^3}{\sqrt{3}}\angle 0^\circ \text{ volts} = 63508.52\angle 0^\circ \text{ volts}$$

Power delivered to load $= \sqrt{3}\, V_R\, I_R \cos\phi_R$

$$\bar{I}_R = \frac{\text{Power delivered to load}}{\sqrt{3}\, V_R \cos\phi_R} = \frac{20\times 10^6}{\sqrt{3}\times 110\times 10^3\times 0.9} = 116.63\ \text{A}$$

$$\therefore \quad \bar{I}_R = 116.63\angle -\cos^{-1}0.9\ \text{A} = 116.63\angle -25.84^\circ\ \text{A}$$

$$\overline{V}_1 = \overline{V}_R + \bar{I}_R\left(\frac{\overline{Z}}{2}\right) = (63508.52\angle 0^\circ) + (116.63\angle -25.84^\circ)\left(\frac{R}{2} + j\frac{X_L}{2}\right)$$

$$= (63508.52 + j0) + \left[(116.63\angle -25.84^\circ)\left(\frac{15.3}{2} + j\frac{38}{2}\right)\right]$$

$$= (63508.52 + j0) + (2388.58\angle 42.22^\circ) = (63508.52 + j0) + (1768.91 + j\,1605.07)$$

$$= 65277.43 + j1605.07 = 65297.16\angle 1.41^\circ \text{ volts}$$

$$\bar{I}_C = \frac{V_1}{Z_C} = \frac{V_1}{0 - jX_C} = \frac{V_1}{-jX_C} = \left(j\frac{1}{X_C}\right)(V_1) = (j3\times 10^{-4})(65297.16\angle 1.41^\circ)$$

$$= (3\times 10^{-4}\angle 90^\circ)(65297.16\angle 1.41^\circ) = 19.58\angle 91.41^\circ\ \text{A}$$

$$\bar{I}_S = \bar{I}_C + \bar{I}_R = (19.58\angle 91.41^\circ) + (116.63\angle -25.84^\circ)$$

$$= (-0.4817 + j19.574) + (104.96 - j\,50.8342)$$

$$= 104.4783 - j\,31.2602 = \mathbf{109.05\angle -16.65^\circ\ A}$$

$$\overline{V}_S = \overline{V}_1 + \bar{I}_S\left(\frac{\overline{Z}}{2}\right) = \overline{V}_1 + \bar{I}_S\left(\frac{R}{2} + j\frac{X_L}{2}\right)$$

$$= (63508.52\angle 0^\circ) + \left[(109.05\angle -16.65^\circ)\left(\frac{15.3}{2} + j\frac{38}{2}\right)\right]$$

$$= (63508.52 + j0) + [(109.05\angle -16.65^\circ)(7.65 + j\,19)]$$

$$= (63508.52 + j0) + [(109.05\angle -16.65^\circ)(20.48\angle 68.06^\circ)]$$

Solving

$$\overline{V}_S = 64901.55 + j1745.64 = 64925.02\angle 1.54^\circ \text{ volts}$$

$$\therefore \quad V_S = 64.925\ \text{kV/phase},\ V_{S\,line} = \sqrt{3}\, V_{S\,ph} = \sqrt{3}\times 64.925$$

$$\therefore \quad V_{S\,line} = 112.45\ \text{kV}$$

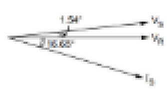


Fig. 2.8.7 (a)

p.f. angle between V_S and I_S

$$\therefore \quad \phi_S = 1.54 + 16.63 = 18.19^\circ$$

p.f. at sending end $= \cos\phi_S = \cos(18.19^\circ) = \mathbf{0.95\ lag}$

$$\%\ \text{voltage regulation} = \frac{V_S - V_R}{V_R}\times 100 = \frac{64.925 - 63.508}{63.508}\times 100 = \mathbf{2.231\ \%}$$

$$\text{Total line losses} = 3I_S^2\frac{R}{2} + 3I_R^2\frac{R}{2} = \frac{3R}{2}(I_S^2 + I_R^2)$$

$$= \frac{3\times 15.3}{2}(109.05^2 + 116.63^2) = 585.09\ \text{kW} = \mathbf{0.58509\ MW}$$

$$\%\ \text{Transmission Efficiency} = \frac{\text{Power delivered to load}}{\text{Power delivered to load} + \text{Losses}}\times 100$$

$$= \frac{20\times 10^6}{20\times 10^6 + 0.58509}\times 100 = \mathbf{97.15\%}$$

```
Enter frequency (Hz): 50
Enter line length (km): 100
Enter resistance per km (ohm/km): 0.153
Enter inductance per km (H/km): 1.21*10^-3
Enter capacitance per km (F/km): 0.00958*10^-6
Enter conductance per km (S/km) [enter 0 if not known]: 0
Select the equivalent model:
1. Nominal π Model
2. Nominal T Model
Enter choice (1 or 2): 2

Line classification: Medium Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End RMS Voltage (V): 110*10^3
Enter load power (W): 20*10^6
Enter power factor (positive for lagging, negative for leading): 0.9

RESULTS FOR MEDIUM LINE (NOMINAL T) MODEL
A = 0.9943 +0.0023j
B = 15.2125 +37.9222j
C = 0.0000 +0.0003j
D = 0.9943 +0.0023j

Receiving end current magnitude: 116.6364 A
Sending end current magnitude  : 109.0464 A

Sending end voltage magnitude    : 66754.42 V
Receiving end voltage magnitude  : 63508.53 V
Voltage Regulation               : 5.72 %
Efficiency                       : 97.16 %
Total losses                     : 195038.0008 W
Sending Power: 6861704.6674 + j2430144.0446 VA
Receiving Power: 6666666.6667 + j3228814.0323 VA
```

Question 4

```
Enter frequency (Hz): 50
Enter line length (km): 150
Enter resistance per km (ohm/km): 0.1
Enter inductance per km (H/km): 1.5927*10^-3
Enter capacitance or susceptance per km (see note) : 9.55*10^-9
Enter conductance per km (S/km) [enter 0 if not known]: 0
Select the equivalent model:
1. Nominal π Model
2. Nominal T Model
Enter choice (1 or 2): 1

Line classification: Medium Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End LINE RMS Voltage (V): 110*10^3
Enter load power (W): 50*10^6
Enter power factor (positive for lagging, negative for leading): 0.8
Input interpreted as C = 9.5500e-09 F/km => B = 3.0002e-06 S/km

RESULTS FOR MEDIUM LINE (NOMINAL Π) MODEL
A = 0.983112 +0.003375j
B = 15.000000 +75.054219j
C = -7.594869e-07 +4.462330e-04j
D = 0.983112 +0.003375j

Receiving end current magnitude (phase): 328.0399 A
Sending end current magnitude   (phase): 306.3794 A
```

```
Sending end phase voltage        : 82898.08 V
Receiving end phase voltage      : 63508.53 V
Sending end line (L-L) voltage   : 143583.69 V
Receiving end line (L-L) voltage: 110000.00 V
Voltage Regulation               : 32.77 %
Efficiency                       : 91.58 %
Total losses                     : 1532834.9411 W

Sending Power (phase): 18199501.6078 + j17715813.7876 VA
Receiving Power(phase): 16666666.6667 + j12500000.0000 VA
Sending Power (3-ph)  : 54598504.8234 + j53147441.3628 VA
Receiving Power (3-ph): 50000000.0000 + j37500000.0000 VA
```

Example 2.8.2 *A 3-phase, 50 Hz, transmission line is 150 km long and deliver 50 MW at 0.8 p.f. lag and at 110 kV. The resistance and reactance of the line per conductor per kilometer are 0.1 ohm and 0.5 ohm respectively. The line charging admittance is 3 × 10-6 mho/km per phase. Compute by apply nominal π method, sending end voltage and current.*

Solution :

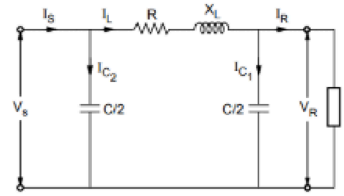


Fig. 2.8.9

Solution : Total $R = 0.1\ \Omega \times 150 = 15\ \Omega$

$$\text{Total } X_L = 0.5\ \Omega \times 150 = 75\ \Omega$$

$$\text{Total } Y_C = 3\times10^{-6}\times150 = 4.5\times10^{-4}\ \text{mho}$$

$$\frac{Y_C}{2} = \frac{4.5\times10^{-4}}{2} = 2.25\times10^{-4}\ \text{mho}$$

$$V_R = \frac{110\times10^3}{\sqrt{3}} = 63508.52\ \text{V}$$

$$= 63.508\ \text{kV}$$

$$\text{Receiving end load current, } I_R = \frac{\text{Power delivered}}{\sqrt{3}\times V_R\times\cos\phi_R} = \frac{50\times10^6}{\sqrt{3}\times110\times10^3\times0.8}$$

$$I_R = 328\ \text{A}$$

$$\bar{I}_R = 328\angle-\cos^{-1}0.8\ \text{A} = 328\angle-36.86^\circ\ \text{A}$$

$$\text{Now,}\quad X_C = \frac{1}{\omega C};\ \omega C = \frac{1}{X_C};\quad \frac{\omega C}{2} = \frac{1}{2X_C} = \frac{Y_C}{2}$$

$$\text{and}\quad \bar{I}_{C_1} = j(2\pi f)\frac{C}{2}\bar{V}_R = j\omega\frac{C}{2}\bar{V}_R$$

$$\therefore\quad \bar{I}_{C_1} = j\frac{Y_C}{2}\bar{V}_R = j\left(2.25\times10^{-4}\right)(63508.52\angle0^\circ)$$

$$= j\ 14.28\ \text{A} = 14.28\angle90^\circ$$

$$\bar{I}_L = \bar{I}_R + \bar{I}_{C_1} = [328\angle-36.86^\circ] + [14.28\angle90^\circ]$$

$$= [262.43 - j\,196.75] + [0 + j\,14.28] = 262.43 - j182.47$$

$$= 319.63\angle-34.81^\circ\ \text{A}$$

$$\text{Now, } \bar{V}_S = \bar{V}_R + \bar{I}_L\bar{Z} = \bar{V}_R + \bar{I}_L[R + jX_L] = [63508.52\angle0^\circ] + [319.63\angle-34.81^\circ][15 + j\,75]$$

$$= [63508.52\angle0^\circ] + [319.63\angle-34.81^\circ][76.48\angle78.69^\circ]$$

$$\text{Solving,}\quad \bar{V}_S = 81128.52 + j16944.26 = 82879\angle11.79^\circ\ \text{volts}$$

$$\therefore\quad V_S = \mathbf{82.87\ kV}$$

$$\text{Line value of sending end voltage} = \sqrt{3}\times82.87 = 143.53\ \text{kV}$$

$$\bar{I}_{C_2} = j\frac{Y_C}{2}\bar{V}_S = j\left(\frac{4.5\times10^{-4}}{2}\right)(82879\angle11.79^\circ)$$

$$= [2.25\times10^{-4}\angle90^\circ][82879\angle11.79^\circ] = 18.64\angle101.79^\circ$$

$$= -3.808 + j\,18.247 = 18.64\angle101.79^\circ\ \text{A}$$

$$\bar{I}_S = \bar{I}_L + \bar{I}_{C_2} = (262.43 - j182.47) + (-3.808 + j\,18.247)$$

$$= 258.62 - j\,164.223 = \mathbf{306.35\angle-32.41^\circ\ A}$$

Question 5

```
Enter frequency (Hz): 50
Enter line length (km): 100
Enter resistance per km (ohm/km): 0.1
Enter inductance per km (H/km): 1.117*10^-3
Enter capacitance or susceptance per km (see note) : 0.9954*10^-12
Enter conductance per km (S/km) [enter 0 if not known]: 0
Select the equivalent model:
1. Nominal π Model
2. Nominal T Model
Enter choice (1 or 2): 1

Line classification: Medium Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End LINE RMS Voltage (V): 66*10^3
Enter load power (W): 20*10^6
Enter power factor (positive for lagging, negative for leading): 0.8
Input interpreted as C = 9.9540e-13 F/km => B = 3.1271e-10 S/km

RESULTS FOR MEDIUM LINE (NOMINAL Π) MODEL
A = 0.999999 +0.000000j
B = 10.000000 +35.091590j
C = -2.444753e-15 +3.127140e-08j
D = 0.999999 +0.000000j

Receiving end current magnitude (phase): 218.6933 A
Sending end current magnitude   (phase): 218.6924 A
```

```
Sending end phase voltage        : 44720.52 V
Receiving end phase voltage      : 38105.12 V
Sending end line (L-L) voltage   : 77458.21 V
Receiving end line (L-L) voltage: 66000.00 V
Voltage Regulation               : 17.36 %
Efficiency                       : 93.31 %
Total losses                     : 478265.9602 W

Sending Power (phase): 7144932.6268 + j6678257.3223 VA
Receiving Power(phase): 6666666.6667 + j5000000.0000 VA
Sending Power (3-ph) : 21434797.8805 + j20034771.9670 VA
Receiving Power (3-ph): 20000000.0000 + j15000000.0000 VA
```

Solution : The nominal π circuit is shown in the Fig. 2.8.12.

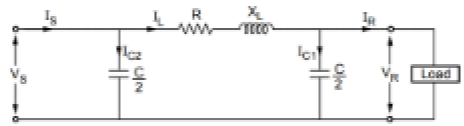


Fig. 2.8.12

$$R = 0.1 \times l = 0.1 \times 100 = 10\ \Omega \qquad \ldots l = 100\text{ km}$$

$$X_L = 2\pi f L \times l = 2\pi \times 50 \times 1.117 \times 10^{-3} \times 100 = 35.1\ \Omega$$

$$Z_C = -jX_C = -j\frac{1}{2\pi f C} = -j\frac{1}{2\pi \times 50 \times 0.9954 \times 10^{-6}}$$

$$= -j\,3197.81\ \Omega/\text{km}$$

$$Z_C = -j\,3197.81 \times (100\text{ km}) = -j\,319781\ \Omega$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{-jX_C} = +j3.127 \times 10^{-6}\text{ mho} \qquad \ldots \frac{-1}{j} = +j$$

$$Y_{C1} = Y_{C2} = \frac{Y_C}{2} = +j1.5635 \times 10^{-6}\text{ mho}$$

$$V_R = 66\text{ kV},\ V_R(\text{ph}) = \frac{66}{\sqrt{3}} = 38.105\text{ kV}$$

$$I_R = \frac{P_R}{\sqrt{3}\,V_R \cos\phi_R} = \frac{20 \times 10^6}{\sqrt{3} \times 66 \times 10^3 \times 0.8} = 218.693\text{ A}$$

$$\bar{I}_R = 218.693 \angle -\cos^{-1}0.8 = 218.693 \angle -36.869^\circ\text{ A}$$

$$\bar{I}_{C1} = \frac{\bar{V}_R}{Z_{C1}} = \bar{V}_R \times Y_{C1} = 38.105 \times 10^3 \angle 0^\circ \times j1.5635 \times 10^{-6}$$

$$= j\,0.0595\text{ A}$$

$$\bar{I}_L = \bar{I}_R + \bar{I}_{C1} = 174.956 - j\,131.21 + j\,0.0595$$

$$= 174.956 - j\,131.15\text{ A} = 218.65 \angle -36.86^\circ\text{ A}$$

$$\bar{V}_S = \bar{V}_R + \bar{I}_L(R + jX_L) = (38.105 \times 10^3 + j0) + \bar{I}_L(10 + j35.1)$$

$$= (38.105 \times 10^3 + j0) + [218.65 \angle -36.86^\circ \times 36.496 \angle 74.09^\circ]$$

$$= 38.105 \times 10^3 + j0 + 6353.66 + j4828$$

$$= 44458.66 + j\,4828\text{ V} = \mathbf{44.72 \angle 6.19^\circ\text{ kV}}$$

$$\therefore \quad V_S(\text{line}) = \sqrt{3} \times 44.72 = 77.45\text{ kV}$$

$$\text{Total line losses} = 3I_L^2 R = 3 \times 218.65^2 \times 10 = 1.434 \times 10^6\text{ W}$$

$$\therefore \text{Transmission efficiency} = \frac{P_R}{P_R + \text{losses}} \times 100$$

$$= \frac{20 \times 10^6}{20 \times 10^6 + 1.434 \times 10^6} \times 100 = \mathbf{93.31\ \%}$$

$$\bar{V}_S = \left[1 + \frac{\bar{Z}\,\bar{Y}}{2}\right]\bar{V}_R + \bar{Z}\,\bar{I}_R$$

$$\text{On no load } I_R = 0 \quad \text{i.e.} \quad \bar{V}_S = \left[1 + \frac{\bar{Z}\,\bar{Y}}{2}\right]\bar{V}_R(\text{NL})$$

$$\therefore \quad \bar{V}_R(\text{NL}) = \frac{\bar{V}_S}{1 + \frac{\bar{Z}\,\bar{Y}}{2}}$$

$$1 + \frac{\bar{Z}\,\bar{Y}}{2} = 1 + \frac{(10 + j35.1)(j3.127 \times 10^{-6})}{2} = 0.999 \angle 8.96 \times 10^{-4}$$

$$\therefore \quad \bar{V}_R(\text{NL}) = \frac{49.72 \angle 6.19^\circ}{0.999 \angle 8.96 \times 10^{-4}} = 44.724 \angle 6.189^\circ\text{ kV}$$

$$\therefore \text{Voltage regulation} = \frac{\bar{V}_R(\text{NL}) - V_R}{V_R} \times 100$$

$$= \frac{44.724 - 38.105}{38.105} \times 100 = \mathbf{17.37\ \%}$$

Question 6

- A 3 phase transmission line of length of 300 km has as an impedance of $(40+j125)\ \Omega$ and admittance of $j10^{-3}$ mho is supplying a load of 50 MW at a voltage of 220 kV with 0.8 pf lag. Find
  1. the sending end voltage (Vs) and
  2. sending current (Is)

- Using Rigorous solution:

$$V_s = V_R\cosh(\gamma l) + I_R\cdot Z_c\sinh(\gamma l)$$

$$I_s = \frac{1}{Z_c}\sinh(\gamma l)\cdot V_R + I_R\cosh(\gamma l)$$

$$V_{R,ph} = \frac{V_{R,L}}{\sqrt{3}} = \frac{220\text{ kV}}{\sqrt{3}} = 127.01\text{ kV}$$

$$\cos\phi_R = 0.8(\text{lag}) \Rightarrow \phi_R = 36.86^\circ$$

$$\text{Load} = 3V_{R,ph}I_{R,ph}\cos\phi_R = 50\times10^6$$

$$\Rightarrow I_{R,ph} = \frac{50\times10^6}{3\times127.01\times10^3\times0.8} = 164.02\text{ A}$$

$$I_{R,ph} = 164.02\angle-36.86^\circ \Leftarrow V_R\text{ Reference}$$

$$Z = (40+j125) = 131.24\angle72.25^\circ\ \Omega$$

$$Y = j10^{-3} = 10^{-3}\angle90^\circ$$

$$Z_c = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{131.24\angle72.75^\circ}{10^{-3}\angle90^\circ}} = 362.27\angle-17.25^\circ\gamma l$$

$$\Rightarrow \gamma l = 0.0558\ (\text{rad})\quad \beta l = 0.3578\ (\text{rad})$$

$$\cosh(\gamma l) = \cosh(\alpha l + j\beta l)\qquad (\text{calculator in radian mode})$$

$$= \cosh(\gamma l)\cosh(\beta l) + j\sinh(\gamma l)\sin(\beta l)$$

$$= 0.9381 + j0.00195 = 0.9383\angle1.19^\circ\ (\text{Calculator in degree mode})$$

$$\sinh(\gamma l) = \sinh(\alpha l + j\beta l)\qquad (\text{calculator in radian mode})$$

$$= \sinh(\gamma l)\cosh(\beta l) + j\cosh(\gamma l)\sin(\beta l)$$

$$= 0.05229 + j0.03507 = 0.3545\angle81.51^\circ\ (\text{calculator in radian mode})$$

$$V_S = V_R\,\text{coshrl} + I_R Z_c\,\text{sinhrl}$$

$$= 137045.35\,|6.20^\circ\text{V}$$

$$= 137.04\angle20^\circ\text{kV}$$

$$I_S = \frac{1}{Z_C}V_R\,\text{sinhVl} + I_R\cosh\gamma l$$

$$= 128.91\angle5.57^\circ\text{A}$$

```
Enter frequency (Hz): 50
Enter line length (km): 300
Enter resistance per km (ohm/km): 0.13333
Enter inductance per km (H/km): 1.326*10^-3
Enter capacitance or susceptance per km (see note) : 3.33*10^-6
Enter conductance per km (S/km) [enter 0 if not known]: 0
Model automatically chosen: LONG (Distributed) line

Line classification: Long Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End LINE RMS Voltage (V): 220*10^3
Enter load power (W): 50*10^6
Enter power factor (positive for lagging, negative for leading): 0.8
Input interpreted as B = 3.3300e-06 S/km => C = 1.0600e-08 F/km

RESULTS FOR LONG LINE (DISTRIBUTED) MODEL
A = 0.938157 +0.019566j
B = 38.349406 +122.649747j
C = -6.570467e-06 +9.783291e-04j
D = 0.938157 +0.019566j

Receiving end current magnitude (phase): 164.0200 A
Sending end current magnitude   (phase): 128.8967 A

Sending end phase voltage       : 137066.13 V
Receiving end phase voltage     : 127017.06 V
Sending end line (L-L) voltage  : 237405.50 V
Receiving end line (L-L) voltage: 220000.00 V
Voltage Regulation              : 15.00 %
Efficiency                      : 95.60 %
Total losses                    : 767144.3781 W

Sending Power (phase): 17433811.0448 + j-2863288.0460 VA
Receiving Power(phase): 16666666.6667 + j12500000.0000 VA
Sending Power (3-ph) : 52301433.1343 + j-8589864.1379 VA
Receiving Power (3-ph): 50000000.0000 + j37500000.0000 VA
```

Question 7:
(cat - 2 qn: lossless transmission line)

```
Enter frequency (Hz): 50
Enter line length (km): 320
Enter resistance per km (ohm/km): 0
Enter inductance per km (H/km): 0.98*10^-3
Enter capacitance or susceptance per km (see note) : 9.93*10^-9
Enter conductance per km (S/km) [enter 0 if not known]: 0
Model automatically chosen: LONG (Distributed) line

Line classification: Long Transmission Line

Is it a 3-phase line?
1 - Yes
0 - No (Single Phase)
Enter choice: 1

Enter Receiving End LINE RMS Voltage (V): 400*10^3
Enter load power (W): 600*10^6
Enter power factor (positive for lagging, negative for leading): 0.8
Input interpreted as C = 9.9300e-09 F/km => B = 3.1196e-06 S/km

RESULTS FOR LONG LINE (DISTRIBUTED) MODEL
A = 0.951227 +0.000000j
B = 0.000000 +96.913353j
C = 0.000000e+00 +9.819894e-04j
D = 0.951227 +0.000000j

Receiving end current magnitude (phase): 1082.5318 A
Sending end current magnitude   (phase): 911.8943 A
```

```
Sending end phase voltage       : 294822.26 V
Receiving end phase voltage     : 230940.11 V
Sending end line (L-L) voltage  : 510647.13 V
Receiving end line (L-L) voltage: 400000.00 V
Voltage Regulation              : 34.21 %
Efficiency                      : 100.00 %
Total losses                    : 0.0000 W
```